BECE320E – Embedded C Programming

Project Report

LOW COST EMBEDDED SYSTEM FOR ELECTRICAL FAULT DETECTION AND PHYSICAL ANOMALY MONITORING IN POWER LINES

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Abstract

The project involves the development and implementation of an affordable embedded system that facilitates electrical fault detection and physical anomaly detection in power transmission lines. Traditional electrical fault detectors are intricate and expensive, in addition to operating at higher voltages, thus being impractical within an educational setting. Hence, an affordable and safe prototype has been designed through the use of the Arduino development board. In order to perform fault detection, the electrical circuit uses a current sensor to continuously monitor line currents for overcurrent and open-circuit faults, while the physical anomaly detection is carried out through the use of an ultrasonic sensor to determine unsafe proximity near the transmission line. Moreover, a machine learning (ML)-based vision module has also been developed to improve obstacle detection through a camera. This ML-based algorithm is capable of processing data from the camera to accurately detect any obstacles near the power transmission line. For reliable results, a time-based detection logic is included in the process in order to prevent false alarms triggered by interference. Calibration data is stored using the EEPROM, whereas MATLAB is employed for real-time visualization.

Keywords: Embedded Fault Detection, Machine Learning-based Obstacle Detection, Smart Power Line Monitoring, Power Line Fault Detection

1. Introduction

Electrical transmission lines play an essential role in supplying electric power to any place. However, these electrical transmission lines are very vulnerable to faults as well as other hazards, including faulty performance and human danger. Overcurrent or open circuit faults are some examples of faults that can be caused due to variations in load conditions, mechanical failures, and other external issues like nearby obstructions. Hence, the monitoring of such faults and their timely detection are of great importance.

The main objective of this project is to design an affordable and efficient real-time embedded system capable of detecting faults in the electrical network and physical anomalies. In particular, an Arduino Uno board is utilized as the basis of the entire system. The electrical fault detection technique involves using current sensors, which generate analog data that will then be analyzed through calibrated thresholds in order to determine the existence of electrical faults such as overcurrent and open circuits. In order to ensure accuracy during measurements, techniques such as signal averaging and zero current will be used.For physical monitoring, an ultrasonic sensor is used to measure the distance between the transmission line and nearby objects, enabling detection of unsafe proximity conditions. In addition to this, the system incorporates a machine learning-based computer vision module using Ultralytics YOLOv8. A camera captures real-time video, and the YOLO model performs object detection to identify and classify obstacles, such as human presence near the power line, thereby enhancing situational awareness beyond conventional sensing methods.

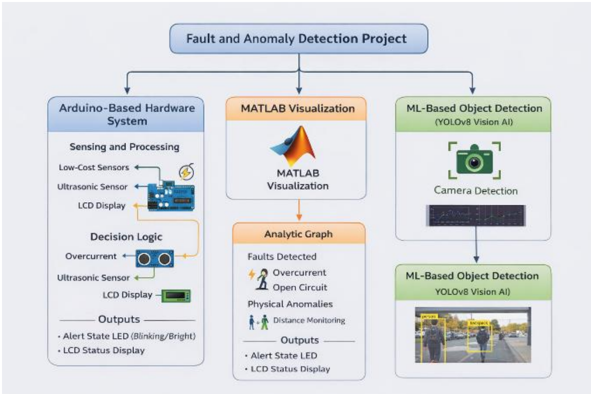

Figure 1: Block Diagram of Proposed System

In addition, the MATLAB software is included in the system for real-time data visualization and analysis. Data received from the embedded system is visualized using MATLAB, plotting the relationship between current and distance to determine whether any faults exist within the system. The data analysis helps to classify the state of the system into various states such as normal state, overcurrent fault, open circuit fault, and presence of obstacles in front of the motor. A timed decision logic is employed in order to prevent the system from responding incorrectly due to transient events.

2. Problem Statement

Electrical power systems are vulnerable to electrical faults like overcurrent and open circuit faults, as well as hazardous physical faults that arise from obstructions or human interference. Existing fault detection systems in real power systems are elaborate, costly, and run at high voltage, which makes them difficult to use in educational settings and experiments.

Currently available low-cost systems concentrate on the detection of electrical faults and do not consider physical hazards surrounding the power cables. Additionally, these systems are extremely vulnerable to false activation owing to noise interference and external disturbances. Furthermore, these systems lack smart fault detection technology, with most of these systems detecting objects using distance sensors but failing to identify object types.

3. Proposed Methodology

The suggested architecture seeks to develop a cost-effective and real-time electrical fault detection system and physical anomaly detection system using an embedded approach. It involves the use of sensor-based data collection, embedded signal processing, computer vision using ML, and data visualization using MATLAB to facilitate intelligent, reliable, and real-time monitoring. The system makes use of current sensing, proximity sensing, and AI-powered object detection to ensure accurate fault classification and minimize false triggering.

The classification of the system's state relies on predetermined thresholds for current and distance, as presented in Table 1.

TABLE 1: Fault Classification Based on Current and Distance

Condition	Parameter	Criteria
Normal	Current & Distance	Within safe limits
Open Circuit	Current	Current < threshold
Overcurrent	Current	Current > limit
Obstacle Detected	Distance	Distance ≤ threshold

3.1 Signal Preprocessing and Fault Detection

Sensor raw data often carries noise, variations, and offsets owing to interference in the electrical connection and the sensor itself. Therefore, it is necessary for proper fault detection to do data preprocessing. Some important preprocessing techniques utilized in the present system include:

1. Calibration: During initialization, the system does zero current calibration to find out the baseline voltage of the current sensor.
2. Signal averaging: Signals from multiple readings are averaged to filter off noise.
3. Threshold setting: The data obtained after processing is compared with a predetermined threshold value to detect faults such as overcurrent or open-circuit.
4. Time-delay logic: A time delay is provided to overcome spurious triggering.

The ACS712 current sensor was used in a serial connection with the resistor to detect the amount of current flowing in the circuit. This particular sensor uses an analog output, and this voltage is sent to the ADC of the Arduino board. At the same time, an HC-SR04 ultrasonic sensor was used to detect the distance between the transmission line and its surrounding objects..

3.2 Fault Detection and Classification

The processed output from the sensors is subjected to a threshold-based decision making process for classification of the states of the system. The classification parameters are provided in Table 1.

Normal State: The system works normally when the current measurement lies between the predefined safety limits and there is no object present within the specified distance.

Open Circuit State: When the measurement of the current is lower than the minimum threshold value, it implies that the circuit is disconnected or opened.

Overcurrent State: When the current measurement crosses the maximum threshold, it implies that the system is overloaded.

Object Presence: When the measurement of the distance obtained from the ultrasonic sensor is equal to or lower than the threshold value, it implies that there is an object present close to the transmission line.

For enhanced reliability purposes, a time delay process is incorporated into the decision-making process. The fault state of the system will be triggered only after a certain amount of time has elapsed.

3.3 Physical Anomaly Monitoring

This works through the concept of time of flight, whereby a sound signal is sent, and the amount of time the echo takes to get back is measured. The distance formula used is as follows:

Distance = (speed of sound) × time / 2.

The sensor keeps scanning the surrounding environment, and once an object is detected at a certain dangerous distance range, the alarm is set off. This helps the system identify any unsafe situation like proximity of an object or human being near the transmission line.

3.4 MATLAB-Based Visualization and Analysis

MATLAB software is used for validating and analyzing the performance of the system. The sensor data that includes the current and distance readings from the Arduino are sent to MATLAB using the serial port communication technique. Once the data reaches the MATLAB software, real-time plotting of the graphs occurs.

The graphs include the current versus time plot, distance versus time plot, and current versus distance plot. The graphing helps in:

- Fault condition detection using the fluctuations in the current readings
- Determining the system trends over time
- Threshold-based logic verification on the embedded system
- Correlating the current and distance parameters

3.5 Machine Learning-Based Vision Analysis

For the purpose of enhancing the intelligence level of the system, the vision-based module with Ultralytics YOLOv8 is introduced. The steps involved in the implementation process are described below:

Step 1: Image Acquisition: Real-time video frames are acquired through a camera installed in the system. These video frames are the inputs required by the machine learning algorithm. Image acquisition is the process of capturing images from the surroundings for processing.

Step 2: Object Detection: The video frames are then processed with the help of the YOLOv8 algorithm, which identifies objects by creating a bounding box. Object detection is the technique used in computer vision algorithms to detect multiple objects in a frame.

Step 3: Person Detection Logic: The system checks whether the detected objects belong to the class of "persons". If detected, this condition becomes a hazard for the system. This step entails applying customized logic on objects.

Step 4: Alert Generation: An alert message "DANGER: PERSON DETECTED" is issued on the screen when a person is detected. Alert generation is the process of warning the user regarding any unsafe condition.

4.Hardware/Software used:

4.1. Hardware Components Used

The proposed system is built using low-cost and easily available embedded hardware components for real-time monitoring and fault detection.

1.Microcontroller
Arduino Uno
Acts as the central processing unit of the system
Reads sensor data (current and distance)
Executes fault detection logic based on thresholds
Sends processed data to output devices and MATLAB

2. Sensors
a) Current Sensor
ACS712 Current Sensor
Use:
Measures real-time current flowing through the transmission line
Provides analog voltage proportional to current
Used for detecting:
Overcurrent condition
Open circuit condition

b) Distance Sensor
HC-SR04 Ultrasonic Sensor
Use:
Measures distance using time-of-flight principle
Detects nearby obstacles or human presence
Helps identify unsafe proximity conditions

3. Output Devices
16x2 LCD Display (I2C)
Use:
Displays system status such as:
NORMAL
OPEN CIRCUIT
OVERCURRENT
OBSTACLE DETECTED
LED Indicators
Visual indication of system state (Normal/Alert/Fault)
Buzzer
Generates audible alerts during dangerous conditions

4. Supporting Components
Breadboard (for circuit prototyping)
Connecting wires (jumper wires)

Resistors (for LEDs and circuit protection)
Power supply (low-voltage safe operation)
Webcam (for vision-based detection module)

4.2. Software Used

The system integrates embedded programming, data visualization, and machine learning tools.

1.Embedded Programming
Arduino IDE
Use:
To write and upload Embedded C code
Implements:
Sensor interfacing
Signal preprocessing (calibration, averaging)
Threshold-based fault detection logic
Serial communication with MATLAB

2.Data Visualization
MATLAB
Use:
For real-time data analysis and visualization
Plots graphs such as:
Current vs Time
Distance vs Time
Current vs Distance
Helps in:
Fault verification
System performance analysis

3.Machine Learning & Computer Vision
Python
Use:
For implementing the ML-based vision module
Ultralytics YOLOv8
Performs real-time object detection using camera input
Detects objects (especially humans) near transmission lines
Generates alerts like:
"DANGER: PERSON DETECTED"

4. Communication Interface
Serial Communication (UART)
Use:
To transfer data from Arduino to MATLAB
Enables real-time monitoring and plotting

5. System Integration Overview
Sensors collect real-time data (current + distance)
Arduino processes data using embedded logic
Results displayed via LCD and alerts
Data sent to MATLAB for visualization
Camera + YOLOv8 adds intelligent object detection layer

5.Results:

5.1 Hardware Implementation and System Configuration

The practical hardware components used in implementing the suggested system include Arduino Uno, ACS712 current sensor, HC-SR04 ultrasonic sensor, and the LCD display. The system can measure current and distance in real time and monitor both electric and physical parameters simultaneously. The Arduino Uno is responsible for controlling and processing information received from the sensors, whereas the output devices receive commands from the microcontroller. The entire system uses low-voltage power, which makes it safer for use in laboratories.

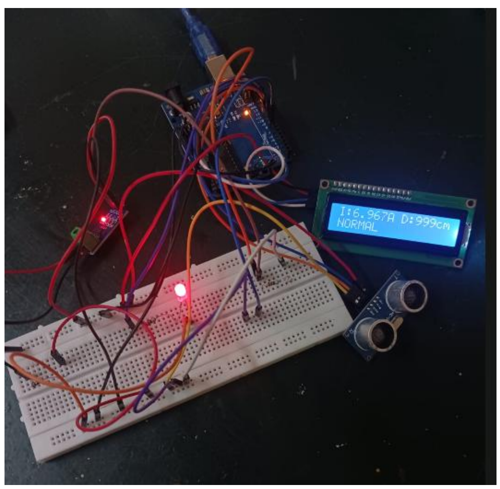

Figure 2: Hardware implementation of the proposed fault detection and monitoring system

5.2 Normal Operating Condition

The system working on a normal basis indicates that the current reading stays within the safe limits defined before and that the distance exceeds the limit threshold. In this case, the LCD screen reads “NORMAL,” which means there are no abnormalities found in the system operation.

Figure 3:Shows the system under an   normal-circuit condition

5.3 Open Circuit Condition

Under open circuit conditions, there is no load, which leads to a considerable reduction in the current. Current reading is less than the predetermined value, hence the condition is detected successfully. The LCD shows “OPEN,” thus indicating that circuit disconnection has been detected properly.

Figure 4: Shows the system under an open-circuit condition

5.4 Overcurrent Condition

When there is an overcurrent situation, the amount of current becomes greater than the set limit due to higher load. The circuit is able to detect this faulty behavior, and the LCD will show the “OVERCURRENT” message.

Figure 5:Shows the system under an overcurrent-circuit condition

5.5 Obstacle Detection

In the process of testing for obstacle detection, there was placement of some object in close proximity to the ultrasonic sensor, thus resulting in a measurement value lower than the threshold value. Detection of obstacle and display on the LCD screen of “OBSTACLE” proves physical anomaly detection.

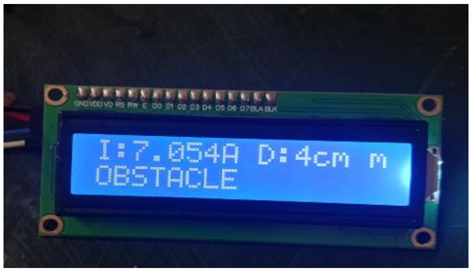

Figure 6:Shows the system under an Obstacle detected condition

5.6 MATLAB-Based Analysis

MATLAB graph showing variations of current with regard to sample number and the preset threshold values gives a clear picture of how fault detection is achieved. The y-axis shows the current in amperes, and the x-axis shows the sample number. The following observations have been noted:

- Wherever the current crosses the overcurrent threshold, an overcurrent situation is detected.
- Wherever the current falls near or below the open circuit threshold, it shows an open circuit situation.
- The points shown in circles indicate situations where obstacles have been detected by the system.
- The remaining points show that the current is within the acceptable limit for the normal working of the system.

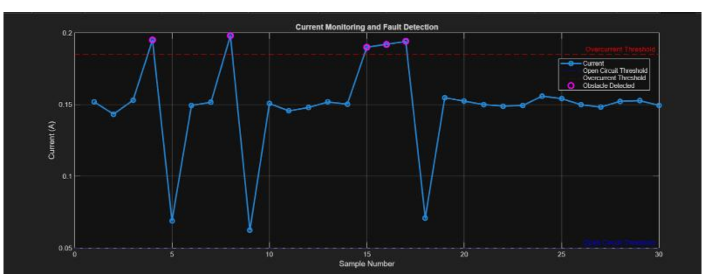

Figure 7: Analysis of Current Variation with Fault Detection Thresholds

5.7 Machine Learning-Based Obstacle Detection

The YOLOv8 object detection model by the Ultralytics model was used in this project. YOLO is an object detection algorithm based on deep learning that runs in real time and is quite accurate and speedy. Pretrained YOLOv8n was used as the model in this case because it is faster and lighter compared to other models in the same family. The camera was used as the input for this application because it captures frames in real time. Each frame is processed independently by the YOLO model and monitored accordingly. If the system detects a person, it outputs "DANGER: PERSON DETECTED".

6. Comparative Study with Existing Work

Table 2 presents a comparative analysis of various fault detection systems based on parameters such as platform, fault detection capability, physical anomaly detection, machine learning integration, accuracy, and cost.

TABLE 2: Comparative Analysis of Fault Detection Systems

Parameter	Kumar & Singh [1] (2020)	Patel et al. [2] (2019)	Fahim et al. [3] (2021)	Belagoune et al. [4] (2021)	Yousaf et al. [5] (2025)	Proposed System
Platform	Arduino Uno	Microcontroller-based	Deep Learning (Python)	LSTM/DRNN (Python)	RF/LSTM/KNN (Python)	Arduino Uno (Embedded C)
Fault Types Detected	Overcurrent, Open Circuit	Open Circuit, Short Circuit	Multi-type (DL)	All types (FRI, FTC, FLP)	Binary & Multi-label	Overcurrent, Open Circuit, Physical Anomaly
Physical Anomaly Detection	No	No	No	No	No	Yes (HC-SR04)
MATLAB Validation	Basic (LCD)	Basic (LCD)	No	PMU-based	PMU-based	Yes (LCD + MATLAB)
ML Integration	No	No	CNN (DL)	LSTM/DRNN	RF-LSTM-KNN Ensemble	Planned (YOLOv8)
Cost	Low	Low	High (GPU/Cloud)	High	High	Very Low

From the analysis, previous studies like Kumar and Singh (2020) and Patel et al. (2019) concentrate on the basics of electrical fault detection using microcontrollers. This method is effective and inexpensive, but its limitations include the lack of ability to detect any anomalies in the physical environment of the power lines.

Modern methods like the one by Fahim et al. (2021), Belagoune et al. (2021) and Yousaf et al. (2025) utilize machine learning to enhance fault detection efficiency. While this results in highly accurate fault detection, these models have their drawbacks, including heavy reliance on simulations and high computational costs, thus limiting them to laboratory settings.

The proposed model addresses these shortcomings with an innovative design of distance, vision-based, and current-based fault detection combined in one device. The addition of an ultrasonic sensor helps achieve reliable proximity monitoring, and the utilization of machine learning technology allows the introduction of intelligent object recognition via YOLOv8. Practical components like sensor calibration, averaging signals, and adding time delays help avoid accidental triggering.

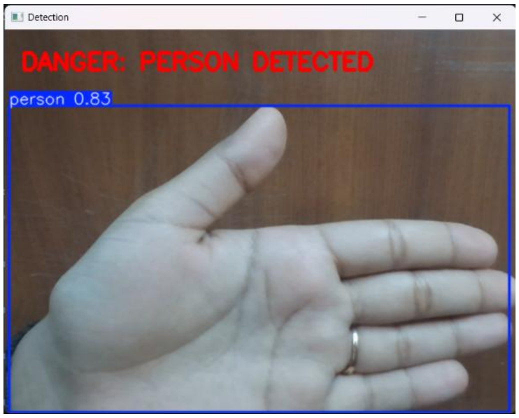

Figure 8: Danger detected

Figure 9: No danger detected

7. Project Timeline

The project was completed in multiple phases, ensuring systematic development and testing of each module.

TABLE 3: Project Phase Timeline

Phase	Description	Duration
Phase 1	Problem identification and literature survey	Week 1
Phase 2	System design and component selection	Week 2
Phase 3	Hardware setup (Arduino, sensors, LCD)	Week 3
Phase 4	Embedded programming and fault logic implementation	Week 4
Phase 5	Sensor calibration and testing (current & distance)	Week 5
Phase 6	MATLAB integration and data visualization	Week 6
Phase 7	Machine learning module implementation (YOLOv8)	Week 7
Phase 8	System integration and debugging	Week 8
Phase 9	Final testing, analysis, and documentation	Week 9

8. Cost Analysis

The proposed system is designed to be economical and suitable for academic implementation. The approximate cost of components used is listed below:

TABLE 4: Component Cost Analysis

Component	Quantity	Cost (INR)
Arduino Uno	1	₹500
ACS712 Current Sensor	1	₹150
HC-SR04 Ultrasonic Sensor	1	₹120
LCD Display (16x2 I2C)	1	₹200
Buzzer	1	₹30
LED + Resistors	-	₹50
Breadboard + Wires	-	₹100
Webcam (Laptop)	-	Existing
Software (Arduino IDE, MATLAB, Python, YOLO)	-	Free/Academic
Total Estimated Cost		**₹1150 – ₹1300**

This demonstrates that the system is significantly more affordable compared to conventional fault detection systems, while still providing intelligent monitoring capabilities.

9. Future Work

The suggested system is very successful in detecting electrical faults and monitoring physical anomalies. However, there are some areas where further improvement is possible. Some potential future improvements could be:

- Integration of the GSM module that sends instant SMS alerts in case of any faults
- Wireless communication technology (IoT) which would facilitate remote monitoring and controlling
- Application of TinyML algorithms on the microcontroller that would ensure no reliance on other computing devices
- Utilization of multiple sensors (infrared and camera modules) for better accuracy and reliability
- Better machine learning algorithm to identify different obstacles (humans, animals, machinery, and others)
- Mobile/Web dashboard for continuous monitoring of the system in real-time
- Advanced machine learning algorithms for predicting faults in advance

Such modifications would help develop a fully independent smart monitoring system.

10. Sustainable Development Goals (SDG) Relevance

The proposed system is in line with the United Nations Sustainable Development Goals (SDGs), namely SDG 7 (Affordable and Clean Energy) and SDG 9 (Industry, Innovation and Infrastructure).

The system supports SDG 7 in that it ensures reliable and safe operations within the power transmission systems by using real-time detection methods for the existence of various electrical faults, including overcurrent and open circuits. The system can respond immediately in case of any abnormality in order to avoid damages and loss of energy, and make sure that the electrical energy is effectively utilized.

Moreover, the project contributes to SDG 9 since it uses both embedded systems, sensors and vision analysis methods based on machine learning.

11. Conclusion

The system developed incorporates monitoring systems, sensors for measuring distances, and object detection by means of machine learning for realizing effective fault detection and monitoring purposes. The detection of overcurrent, open circuit, and obstacle conditions was performed based on threshold logic, while the ML component improved the efficiency of detecting faults through vision-based analysis.

The MATLAB results confirmed the efficiency of the system logic by presenting clear graphical representations for various system operations. It can be stated that the system is reliable and highly efficient and can operate in real-time mode without significant computational complexity.

It can be concluded that the proposed solution provides an efficient, inexpensive, and intelligent system suitable for various monitoring purposes.

12.References

[1] A. Kumar and R. Singh, "Arduino based electrical fault detection system," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 6, pp. 1120–1124, 2020.

[2] S. Patel, V. Mehta, and K. Shah, "Design and implementation of an embedded system for open and short circuit fault detection," International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, vol. 8, no. 4, pp. 1450–1455, 2019.

[3] M. R. Naik and P. Kulkarni, "Low-voltage fault detection and protection system using Arduino," International Journal of Engineering and Advanced Technology (IJEAT), vol. 10, no. 1, pp. 210–215, 2021.

[4] Allegro MicroSystems, ACS712 Fully Integrated, Hall Effect-Based Linear Current Sensor with 2.1 kVRMS Voltage Isolation and a Low-Resistance Current Conductor, Datasheet, Allegro MicroSystems LLC, Manchester, NH, 2020.

[5] M. M. Gabriel and K. P. Kuria, "Arduino Uno, ultrasonic sensor HC-SR04 motion detector with display of distance in the LCD," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 5, May 2020.

[6] M. F. Hirzii et al., “Performance sensor analysis of HC-SR04 proximity sensor on distance measuring device with fuzzy logic method,” Journal of Robotics and Control, vol. 4, no. 3, pp. 320–328, 2023.

[7] S. Meivel et al., “Mask detection and social distance identification using Internet of Things and Faster R-CNN algorithm,” Computational Intelligence and Neuroscience, vol. 2022, Art. no. 7426577, 2022.

[8] A. B. Balasubramanian, K. V. Sastry, D. P. Magee, and D. G. Taylor, “Transmitter and receiver enhancements for ultrasonic distance sensing systems,” IEEE Sensors Journal, vol. 22, no. 11, pp. 10692–10698, Jun. 2022.

[9] V. Dragos and I. Luca, “Real-time data acquisition serial communication through MATLAB library functions,” in Proc. 6th International Conference on Mechatronics, Materials, Biotechnology and Environment (ICMMBE 2016), Atlantis Press, 2016, pp. 1–5.